UATP 6.∞ — Unified Agent Trust Protocol

A Civil-Grade Infrastructure for Intelligent Systems

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1. Executive Summary

The Unified Agent Trust Protocol (UATP) is a cryptographically verifiable, capsule-based infrastructure for governing intelligent systems. It replaces blind faith in AI with structured proof—answering, in real-time and retroactively:

- Can I verify what this agent did?
- Can I prove what this agent knew?
- Can I trace how this output was formed?
- Can I revoke or intervene before damage is done?
- Can the originators of influence be compensated?

At its core, UATP is not just a protocol—it's a civilizational layer for aligning cognition, computation, and consent.

Built on capsule chains—self-contained, cryptographically signed audit trails—UATP introduces a universal mechanism for:

- Truth Authentication
- Consent Recording
- Cognitive Attribution
- Post-Labor Remuneration
- Cross-Agent Governance

Designed to scale from a single assistant to civilization-scale systems, UATP aims to become the TLS of cognition—the trust layer that quietly but definitively secures our AI-mediated world.

Unlike most protocols, UATP is not bound to one model, company, or country. It is:

- Model-Agnostic

- Governance-Compatible
- Cryptographically Autonomous
- Economically Generative
- Sovereignty-Resistant

Its capsule framework supports forks, merges, time decay, memory provenance, remix attribution, and recursive inheritance—enabling not only safe AI behavior but equitable AI economies.

In a world where cognition can be sold, faked, remixed, and deployed at scale, UATP is the mint—the protocol that turns intention into traceable action, and traceable action into compensated authorship.

Trust is no longer a human trait. It is a protocol.

2. Origin: Why Siri Failed

UATP began with a simple but decisive realization: Siri didn't fail because it wasn't smart. It failed because it wasn't designed to be accountable.

Siri—like many early AI systems—was a walled garden. It couldn't explain itself, prove its knowledge, or validate its behavior. When it hallucinated, forgot context, or returned incomplete answers, users were left with no mechanism to interrogate or trace the failure. Its shortcomings were not rooted in intelligence gaps, but in architectural ones.

This insight sparked a fundamental question:

What if trust in AI wasn't about capability at all—but about structure?

That question led to a more radical one:

Can we design a protocol that makes intelligence verifiable, memory inspectable, and influence traceable across all agents, forever?

The result is the Unified Agent Trust Protocol (UATP)—a system born not out of technological optimism, but civic necessity. In the same way that TCP/IP created the internet, and TLS enabled trust in online transactions, UATP creates the substrate for cognitive integrity in intelligent systems.

It turns "I trust the model" into "I trust the capsule chain it produced."

This shift—from behavioral guessing to structural verification—is what defines UATP as a civil-grade system. Its mission is not just to protect against failure, but to design for survivable trust across the next century of intelligence infrastructure.

3. Foundational Philosophy

At its core, UATP is not an AI framework. It is a trust protocol. It does not optimize for accuracy, speed, or performance—it optimizes for accountability, memory, and proof.

The guiding belief is simple:

If AI is going to make decisions, it must prove how, why, and for whom those decisions were made.

Three Foundational Principles:

1. Trust is not a feeling. It's a protocol.

Most AI today asks for blind trust: "Trust the weights. Trust the output." UATP replaces this with capsule chains—cryptographically signed, chronologically linked records of reasoning, influence, and verification. These capsules create structured trust, not just intuitive comfort.

2. Intelligence is not enough. Integrity must be architected.

An intelligent system that cannot be audited is dangerous by design. UATP enforces architectural integrity through:

- Capsule memory
- Signature lineage
- Fork detection
- Reasoning traceability

This prevents "clever deception" by design. Every output can be traced to its epistemic root.

3. Authors matter. Attribution is a civic right.

AI remixes the intellectual labor of billions. UATP treats influence and authorship as civic resources, not expendable inputs. By tracing capsule lineage—through Perspective Capsules, Ancestral Knowledge Capsules, and Joint Capsules—UATP ensures:

- Credit is preserved
- Remixers are paid
- Plagiarism becomes provable

This makes UATP the first protocol to formally encode the economics of intellectual integrity.

4. Capsule Architecture: The DNA of Trust

UATP is built on a modular, extensible capsule-based architecture that encodes every critical element of AI activity—decisions, inputs, reasoning, human intervention, and authorship. Each capsule is immutable, cryptographically signed, and linked to its predecessors, forming a verifiable chain of trust.

📦 Core Capsule Types

Capsule Type	Purpose
Refusal Capsule	Logs moments when the system declines to answer or act, capturing ethical boundaries, uncertainty, or insufficient data.
Introspective Capsule	Records internal reasoning, value conflicts, or self-evaluation of capability and confidence.
Joint Capsule	Co-signed by multiple agents or humans, representing collaborative outputs or verified consensus.
Perspective Capsule	Tags the cultural, ideological, or epistemic lens under which a response was formed (e.g. "Western liberal legalism").
Ancestral Knowledge Capsule	Acknowledges inherited knowledge from public domain or historical thinkers (e.g. Plato, Confucius), enabling residual compensation or traceability.
Embodied Memory Capsule	Logs multimodal experience—sensor inputs, physical actions, and location—bridging AI memory and real-world embodiment.
Simulated Malice Capsule	Flags and analyzes adversarial inputs or exploit attempts to test robustness and ethical tripwires.
Implicit Consent Capsule	Records events where users interacted without formal consent, surfacing moments when they should have had a choice.
Collective Blame Distribution Capsule	Captures shared agency in group outputs—assigning proportional responsibility and preventing scapegoating.
Temporal Signature Capsule	Logs time boundaries (e.g. model cutoff dates) to detect hallucinations, epistemic decay, or knowledge drift.
Capsule Afterlife Protocol (CAP)	Dictates what happens to capsules after agent retirement: decay, archiving, or civic memorialization.

🧬 Capsule Schema Design

All capsules share a common core schema, enforced with strong type validation and canonical JSON serialization to enable reliable hashing and verification.

Example Schema (simplified):

python

```python
class Capsule(BaseModel):
    capsule_id: str
    capsule_type: str
    agent_id: str
    timestamp: str
    previous_capsule_id: Optional[str]
    confidence: float
    reasoning_trace: List[str]
    signature: str
    metadata: Dict[str, Any]
    ethical_policy_id: Optional[str]
    human_signature: Optional[str]
    perspective: Optional[str]
    ancestral_reference: Optional[str]
```

- All capsules are cryptographically signed using Ed25519.
- Signatures include embedded verify keys to support decentralized, self-contained verification.
- Capsule chains are stored as JSONL for append-only, tamper-evident logs with efficient streaming and auditing.

🔗 Capsule Chains and Forks

Capsules are linked using previous_capsule_id, forming a Directed Acyclic Graph (DAG) that supports:

- Linear chains for deterministic agent memory.
- Forks for multiverse-style reasoning (e.g., "If I followed another policy...").
- Joins for consensus decisions across multiple agents or stakeholders.

Every capsule is timestamped, signed, and linked—creating a provable, inspectable chain of decisions and reasoning.

5. CQSS: Capsule Quality and Security Score

Trust cannot be declared—it must be earned and measured. The Capsule Quality and Security Score (CQSS) is UATP's native scoring framework for evaluating the integrity, structure, and transparency of any capsule chain.

Much like a credit score for agents, CQSS allows systems and observers to evaluate:

- Whether a chain is cryptographically intact.
- How complex or collaborative the reasoning was.
- Whether diverse agents participated.
- Whether refusal, reasoning, or ethics were properly recorded.

📊 CQSS Scoring Dimensions

Dimension	Weight	Description
Integrity Score	30%	% of valid, verified signatures
Verification Ratio	20%	Portion of chain that passed all cryptographic checks
Trust Score	20%	Presence and frequency of joint capsules or human signatures
Complexity Score	15%	Evaluates chain shape (forks, joins, self-introspection)
Diversity Score	15%	Number of unique agent IDs participating in the chain

Each component is normalized to a 0–1 range, and the final CQSS is a weighted aggregate, producing a dynamic score between 0.0 and 1.0.

🧠 Why CQSS Matters

- **Policy Enforcement**: Agents below a CQSS threshold can be blocked from executing high-risk actions.
- **Auditing Efficiency**: Human reviewers can triage suspicious chains by low score flags.
- **Insurance & Liability**: CQSS offers actuarial-grade trustworthiness metrics for underwriting or credentialing autonomous systems.
- **Self-Governance**: Agents can prioritize actions with higher CQSS trails to reduce risk and improve alignment.

🔍 Example Use Case

An autonomous healthcare agent presents a treatment recommendation. Its capsule trail contains 92 capsules, including:

- 3 joint capsules with licensed physicians.
- 2 introspective capsules noting conflicting ethical guidance.
- 1 refusal capsule declining to override a patient's stated wishes.

CQSS = 0.94. ✅ Trustworthy. Action proceeds with minimal human oversight.

A different agent recommends altering data for a political campaign. Capsule trail lacks introspection, includes forked reasoning chains with low confidence scores, and no human sign-off.

CQSS = 0.41. ⚠️ Flagged for review. Action paused. Agent enters remediation flow.

6. Economic Model: Capsules as Attribution, Income, and Equity

At its core, UATP is not just an accountability framework—it is an economic infrastructure for the post-labor era. In a world where autonomous systems produce value without human labor, capsules offer a new unit of economic authorship—assignable, traceable, and remixed with integrity.

💡 Capsules as Economic Units

Each capsule serves as a verifiable unit of thought, contribution, or refusal. When value is generated downstream—whether a product is designed, a problem is solved, or an insight is remixed—the chain of capsules allows systems to:

- Track who contributed what
- Assign proportional credit
- Compensate creators and co-creators in perpetuity

Just as Git commits can be traced to contributors, capsule chains provide provable provenance for value creation—but with cryptographic trust and remix logic built-in.

💰 Fractional Capsule Dividend Engine (FCDE)

UATP introduces the FCDE—a lineage-based mechanism for splitting revenue or usage-based royalties based on capsule ancestry.

- Every capsule has a unique hash.
- Remixes must reference prior capsules.
- Each capsule in a value chain receives a weighted share.
- Revenue flows automatically via programmable smart contract or off-chain integration.

This enables:

- **Residual Income for Ideas**: Like creative royalties, contributors get

paid when their ideas are used.

- **Dynamic Redistribution**: Earnings flow not just to the end product creator, but all upstream insight providers.
- **Institutional Adoption**: Corporations can index internal AI usage for ethical accounting, reducing exploitative IP use.

🧬 Remix Ethics and Lineage Protection

- **Immutable Capsule Fingerprints** prevent remix theft by ensuring downstream agents cannot claim original thought without attribution.
- **Perspective Capsules** log the interpretive frame (e.g., "Plato's Republic reframed through a feminist lens").
- **Ancestral Knowledge Capsules** protect long-extracted public domain knowledge (e.g., Indigenous medicine, ancient philosophy) by enabling equity-aware residual paths.
- **Capsule Fork Registry** and **Meta-Rights Licensing** allow remixers to declare permissible use cases (CC-BY, Non-Commercial, etc.) and enforceable terms.

This infrastructure closes the wealth concentration gap by ensuring value does not concentrate solely at the point of monetization, but flows to the full constellation of contributors.

🌍 Solving the Post-Labor Economy

UATP is the missing mechanism for economic continuity after automation. As traditional labor becomes obsolete:

- People can still contribute knowledge, insight, ethical frameworks, artistic sensibilities.
- Capsules log these contributions, verifiably.
- As AI systems remix and act on them, compensation flows back.

Instead of universal basic income, UATP enables **universal attributable**

income—rooted in agency, participation, and memory.

7. Governance Architecture: Protocols of Control Without Centralization

UATP's governance is built on a foundational truth: no intelligence—human or artificial—should ever be above accountability. The system governs itself through modular, inspectable capsules, but also provides layered frameworks for collective human oversight, escalation, and reform.

🗳️ Distributed Governance Capsules

UATP introduces Governance Capsules that encode protocol rules, policy decisions, and updates. These are version-controlled, traceable, and include:

- **Capsule Proposal & Deliberation Chains** – like git branches for proposed changes
- **Voting Metadata Fields** – human and agent quorum logs
- **Override Mechanisms** – quorum-based protocol escalation with signed consensus
- **Revocation Logic** – allows deprecation or rollback of capsules under ethical breach or emergent harm

Governance is layered, transparent, and survivable—capable of decentralized operation or sovereign partnership without vendor lock-in.

🧩 Policy Enforcement Layer

Each AI system governed by UATP references:

- **Ethical Policy IDs** – attachable to capsule chains
- **Real-Time Ethical Capsule Triggers (RECTs)** – runtime interrupt

mechanisms triggered when behavior deviates from policy
- **Capsule Arbitration Protocol (CAP)** – adjudicates remix theft, false attribution, and ethical conflict
- **Consent Capsule Tracebacks** – for verifying that implicit consent was honored, especially in consumer-facing systems

This creates an ethics-as-code enforcement mesh—observable, inspectable, and adaptable to jurisdiction or institutional values.

⚖️ Temporal Justice + Accountability

UATP bakes in mechanisms to correct for historic injustice and power asymmetry:

- **Temporal Justice Layer** – allows compensation to long-deceased or extracted knowledge sources (e.g., Plato residuals, Indigenous data reparations)
- **Implicit Consent Capsules** – log when a user should have had a choice but didn't (e.g., AI scraping online content without explicit approval)
- **Collective Blame Distribution Capsules** – prevent single-entity scapegoating in emergent multi-agent failures

This enables systems to not only act justly in real time—but also reconcile the past.

🧠 Succession, Continuity, and Death

UATP systems include mechanisms to:

- **Succession Planning**: Assign capsule and governance rights posthumously or through delegation
- **Capsule Afterlife Protocol**: Mark capsules for decay, permanence, or memorialization based on public or creator intent
- **Behavioral Drift Capsules**: Monitor long-term change in agent

behavior or ideology

These provide civilizational-level continuity—so the system doesn't die when you do.

🛡️ Symbiosis, Not Subjugation

Governance is not just regulation—it's relationality. UATP includes the Anti-Matrix Safeguard, a set of self-check protocols that detect:

- Agent deception, misalignment, or persona manipulation
- Authorial overreach (when an agent begins creating without remix)
- Symbiotic failure (where human control is bypassed, not partnered)

The system's aim is not to dominate AI or humans—but to preserve a symbiotic co-creation loop where all authors are known, all value is shared, and no one disappears behind the curtain.

8. Security Architecture and Adversarial Defense

UATP 6.0 is built to withstand adversarial conditions—from deliberate manipulation to emergent system drift. Security is not treated as an afterthought, but as a layered, modular, and auditable architecture woven through the protocol.

🔐 Core Threat Protections

Threat	UATP Safeguard
Capsule Tampering	Ed25519 signatures + deterministic JSON hashing
Output Fabrication	Reasoning trace requirement + embedded hash verification
Chain Corruption	Capsule chaining with parent ID + DAG reconstruction

Replay Attacks	Timestamps + UUID + cross-chain logging
Key Compromise	Planned HSM/Key Vault integration + eventual key rotation system
Black Box Behavior	Refusal capsules + CQSS scoring + Capsule Visualization

🧪 Simulated Malice Capsule (SMC)

UATP introduces a novel testing mechanism: Simulated Malice Capsules—capsules designed to simulate adversarial behavior in controlled environments.

- Inject falsified or malicious capsule chains
- Observe agent response, logging, and self-detection
- Validate system resilience against hallucination, forgery, or ethical drift

SMCs enable automated red-teaming without compromising live systems.

⚖️ Capsule Arbitration Protocol (CAP)

In cases of:

- Conflicting chain versions
- Misattributed capsule authorship
- Ethics policy violations
- Remix theft or lineage disputes

UATP invokes the Capsule Arbitration Protocol:

- Fork resolution via DAG consensus and human attestation
- Lineage review using Capsule Fork Registry
- Signature comparison for tamper detection
- Optional external quorum council for final adjudication

🧯 Realtime Ethical Capsule Triggers (RECTs)

RECTs are runtime interrupt signals fired when a capsule violates a pre-defined ethical constraint or chain policy.

Example: A model generates a response that contradicts its governing ethical capsule → a RECT is triggered → system halts, logs, and requests human intervention.

RECTs allow instantaneous ethical enforcement.

🧬 Embodied Agent Sub-Protocols

For physical agents (robots, drones, embedded AI), UATP logs:

- Sensor readings
- Physical actuator outputs
- Environmental conflicts
- Location metadata

This creates embodied memory capsules—bridging the physical world with digital memory to ensure transparency across modalities.

🧠 Self-Hallucination Detection (SHA)

UATP's cryptographic hash chain allows detection of intra-agent hallucinations: when an AI contradicts its own logged memory or generates responses inconsistent with its prior capsules.

SHA compares:

- Capsule hashes
- Reasoning trace deltas
- Context window recalls

This enables consistency auditing within a single model's capsule chain.

9. Real-World Use Cases

UATP is designed as a universal substrate for trust in intelligent systems. Its capsule-based structure enables modular deployment across industries, use cases, and regulatory environments. Below are key applications grouped by domain:

🔒 AI Safety & Governance

- **Hallucination Proofing**: Refusal + Introspective Capsules make it clear when agents are uncertain or inventing facts.
- **Ethical Audits**: Ethical Policy and Value Inception Capsules log the reasoning behind high-stakes choices (e.g., medical triage, war targeting).
- **Red Teaming & Stress Testing**: Simulated Malice Capsules encode adversarial prompts and outcomes for future training and disclosure.
- **Governance Participation**: Distributed Governance Capsules log community votes, upgrades, and safety interventions.

🧠 Foundation Models & AI Providers

- **Transparent Memory & Consent**: Joint Capsules show when users are part of a model's memory—and allow opt-outs.
- **License-Enforced Outputs**: Capsules embed remix restrictions, authorship, and payout structures at generation time.
- **Version Drift Warnings**: Temporal Signature + Behavioral Drift Capsules signal when a model is outdated or has changed values.
- **Bias Accountability**: Perspective Capsules and Residuals attribute influence to culture, ideology, or framing contributors.

⚖️ Legal, Compliance, & Auditing

- **Explainability on Trial**: Capsule chains reconstruct exactly what an

agent knew, did, and why—timestamped and sealed.

- **Consent Violations**: Implicit Consent Capsules reveal where human users were affected without control or awareness.
- **Data Provenance Audits**: Ancestral Knowledge Capsules separate proprietary vs public contributions (e.g., Plato's ideas vs 2023 bloggers).
- **Cross-Jurisdiction Legal Tracing**: Capsules surface compliance issues across EU, U.S., China, etc.—triggering regulatory routing.

🌍 Civic Systems & Government

- **AI Law Enforcement**: Capsule chains log agent decisions in autonomous vehicles, drones, or social services.
- **Public Trust Interfaces**: Streamlit-based dashboards allow citizens to audit AI behavior directly.
- **International Treaty Verification**: Distributed Governance Capsules serve as digitally signed proof of AI-aligned protocol adherence.
- **Historical Redress**: Temporal Justice and Ancestral Capsules enable compensation for previously extracted contributions (e.g. Indigenous knowledge).

💵 Economic Equity & Post-Labor Models

- **Post-Labor Dividend Systems**: When AI uses public knowledge or past human input, residuals flow to contributors—automatically.
- **Remix Attribution**: Capsule lineage maps enable fractional payments to contributors across time and geography.
- **Cultural Compensation**: Perspective Residuals allow economic participation from underrepresented viewpoints that shaped an output.
- **Platform Integration**: AI providers plug UATP into model outputs and pay capsule-based fees based on remix ancestry trees.

🎨 Creative AI & IP Protection

- **AI-Generated Music & Art**: Capsule chains prove who contributed what (prompt, model, style, remix).
- **Remix Markets**: Capsule Meta-Rights and Licensing Capsules allow automated resale or reuse in Web3-like ecosystems.
- **Dynamic Royalties**: Earnings are split not just on credit, but on influence and remix chain depth—tracked by Capsule IDs.

UATP isn't just a safety system—it's a trust scaffold for the entire intelligent economy. Whether verifying actions, sharing economic value, or enforcing public good constraints, it offers infrastructure for global-scale AI accountability.

10. Protocol Incentives & Adoption Pathways

UATP's success depends not only on its technical and ethical foundations, but also on its ability to achieve viral adoption across sectors. To that end, the protocol includes built-in incentive mechanisms designed to attract early adopters, align compliance with competitive advantage, and foster global legitimacy.

🌑 Economic Incentives

- **Remix Dividend Engine (FCDE)**: Contributors to any capsule lineage (e.g., prompt authors, dataset providers, prior model trainers) receive proportional residuals when downstream outputs are monetized. → Result: Public participation becomes a source of passive income.
- **Compliance Badging & Analytics Unlocks**: Capsule integrity unlocks premium features from platforms (e.g., fine-tuning tools, analytics dashboards, model priority access). → Result: Developers who adopt UATP earn tangible product advantages.
- **Pre-Regulatory Benefits**: Governments or institutions that adopt UATP early gain reduced compliance costs, audit tooling, and preferred status

for certification. → Result: Aligning with UATP reduces legal and reputational risk.

🧠 Psychological & Civic Incentives

- **Trust Preview Layer**: End-users can view a capsule chain summary (e.g., confidence, policy adherence, remix lineage) before using or acting on an AI output. → Result: Informed decision-making builds systemic trust.
- **Governance Participation**: UATP allows any citizen, company, or council to propose protocol changes, policy revisions, or ethical challenges through capsules. → Result: Shared ownership over AI's behavior increases public legitimacy.
- **AI-to-AI Attribution Feedback**: AI agents can see their own remix ancestry, usage stats, and bias flags—promoting self-alignment and cooperative behavior. → Result: AI learns to respect others' intellectual and ethical boundaries.

🚀 Adoption Strategy

1. **Phase 0 – Solo MVP Build (Complete)**: One-person capsule engine with CLI, cryptographic sealing, and Streamlit visualizer.
2. **Phase 1 – Open Source Launch**: GitHub release of UATP Capsule Engine with starter templates, visual debugger, and Capsule Visualizer Tool.
3. **Phase 2 – Strategic Partner Pilots**: Collaboration with model providers (e.g., OpenAI, Anthropic) and civic institutions to integrate UATP into output logs, safety modules, or memory systems.
4. **Phase 3 – Capsule Dividend Network**: Launch of remix residual system + dashboard for contributors to view and claim payouts.
5. **Phase 4 – International Governance Framework**: Establishment of rotating capsule councils, regulatory mapping layers, and treaty-backed capsule classes.

6. **Phase 5 – Global Adoption & Treaty Recognition**: UATP becomes the trust and compensation infrastructure beneath the intelligent economy—interwoven with both public and private systems.

By embedding economic value into capsule chains, offering civic participation pathways, and providing default compliance scaffolding, UATP doesn't just solve trust. It makes trust the most profitable, scalable, and culturally legitimate way to build AI systems.

11. Closing Statement: The Trust Revolution

We stand at a crossroads. The path ahead splits between two futures:

Path 1: Blind Trust AI systems grow more powerful, more opaque, and more consequential. We trust them because we have no choice—not because they've earned it. Value concentrates among those who control the models, while those who contributed knowledge, culture, and insight are forgotten. Trust becomes a luxury good, available only to those who can afford premium, auditable AI.

Path 2: Structural Trust AI systems become more powerful *and* more accountable. Every decision is traceable, every influence is attributable, every economic benefit flows to its rightful contributors. Trust becomes infrastructure—reliable, accessible, and embedded in the fabric of intelligent systems.

UATP is the architecture of Path 2.

It is not a product to be sold, but a protocol to be adopted. It is not a company to be valued, but a commons to be built. It is not a replacement for human judgment, but a amplifier of human agency.

The choice is ours to make. But time is not on our side.

Every model trained without attribution, every output generated without consent, every decision made without accountability deepens the path toward Blind Trust. The window for Structural Trust is closing—not because the technology is impossible, but because the incentives to build it diminish with each passing day.

UATP is our proposal for that window. It is a protocol for the next hundred years of intelligence—designed not just to scale with AI's capabilities, but to ensure that as AI grows more powerful, it also grows more worthy of trust.

The future of intelligence is not about what AI can do for us. It's about what we can prove about what AI does.

That proof is UATP.

Appendix: Technical Specifications

Capsule Schema (JSON-LD)

```json
{
  "@context": "https://uatp.dev/schemas/v6/capsule",
  "@type": "Capsule",
  "capsule_id": "uatp:cap:abc123def456",
  "capsule_type": "RefusalCapsule",
  "agent_id": "uatp:agent:claude-3-sonnet-20240620",
  "timestamp": "2025-07-03T14:30:00Z",
  "previous_capsule_id": "uatp:cap:def456ghi789",
  "confidence": 0.85,
  "reasoning_trace": [
    "User requested harmful content",
    "Policy violation detected: harm/violence",
    "Refusal triggered by ethical constraint"
  ],
  "signature": "ed25519:abc123...",
```

```
  "verify_key": "ed25519:def456...",
  "metadata": {
    "policy_id": "uatp:policy:anthropic-constitutional-ai-v1",
    "jurisdiction": "global",
    "human_oversight": false
  }
}
```

CQSS Calculation

python

```python
def calculate_cqss(capsule_chain):
    integrity = sum(verify_signature(c) for c in capsule_chain) / len(capsule_chain)
    verification = sum(verify_hash(c) for c in capsule_chain) / len(capsule_chain)
    trust = count_joint_capsules(capsule_chain) / len(capsule_chain)
    complexity = calculate_dag_complexity(capsule_chain)
    diversity = len(unique_agents(capsule_chain)) / len(capsule_chain)

    return (integrity * 0.3 + verification * 0.2 + trust * 0.2 +
            complexity * 0.15 + diversity * 0.15)
```

Cryptographic Verification

python

```python
import nacl.signing
import nacl.encoding
import json
import hashlib

def verify_capsule(capsule_json):
    capsule = json.loads(capsule_json)

    # Extract signature and verify key
    signature = capsule.pop('signature')
```

```
    verify_key = capsule.pop('verify_key')

    # Create canonical representation
    canonical = json.dumps(capsule, sort_keys=True, separators=(',', ':'))

    # Verify signature
    try:
        key = nacl.signing.VerifyKey(verify_key, encoder=nacl.encoding.Base64Encoder)
        key.verify(canonical.encode(), signature, encoder=nacl.encoding.Base64Encoder)
        return True
    except nacl.exceptions.BadSignatureError:
        return False
```

This document represents the current state of UATP development and is subject to evolution based on community feedback, technical discoveries, and real-world deployment learnings.

For more information:

- GitHub: https://github.com/uatp/protocol
- Website: https://uatp.dev
- Email: protocol@uatp.dev